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Article

Climate Change Effects on the Crop Yield and Its Variability in Telangana, India

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Abstract

This study has examined the effects of climatic factors on mean yields and yield variability of four primary crops (rice, cotton, jowar and groundnut) in Telangana state by applying the Just and Pope production function over a period of 1956–2015. Using the three-stage feasible generalised least squares estimation procedure, we have estimated the production function of four crops. The empirical results have revealed that the effects of changes in climatic factors vary among crops under study. Maximum temperature has a significant adverse effect on rice, cotton and groundnut yields. Minimum temperature has a substantial positive effect on rice, cotton and groundnut. Further, rainfall is adversely related to cotton and groundnut yields. Maximum temperature has appeared as a risk-reducing factor for all study crops while minimum temperature as a risk-enhancing factor for rice, cotton and jowar. Lastly, rainfall has been found as a risk-enhancing factor for rice and groundnut whereas it is a risk-reducing factor for jowar and cotton. Results from the study have important implications on how Telangana's farming sector will adapt to climate variability and change for sustainable agricultural development.

Keywords

Climate change, crop yields, variability, panel data, Telangana, India

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Introduction

Climate change due to natural and anthropogenic activities is considered to be one of the serious environmental issues in the world (Stern, 2006). Pieces of evidence of the changes in rainfall, temperature and extreme climatic events have been established on a more systematic as well as scientific basis (IPCC, 2007; 2014). Extreme climatic events, namely, floods, cyclones, droughts, heat waves and climate change harm global socio-economic, biophysical and ecological systems (IPCC, 2014). More specifically the agricultural sector is considered to be more vulnerable to climate change (Hasan et al., 2016). Since climatic factors such as rainfall and temperature serve as important direct inputs to the crop sector, any change and variability in these variables are inevitable to have a significant effect on crop yields (Barnwal & Kotani, 2013). There is also a generous concern about projected future changes in climatic factors due to rapidly increasing concentrations of greenhouse gases (Aggarwal, 2003) and expected changes in these climate variables would have a direct or indirect impact on food production (Krupa, 2003). Therefore, this area has captured the extensive attention of policy makers and researchers (see Mall et al., 2006; Lobell et al., 2011). Such attention can be more in the less developed countries like India where agricultural production is largely affected by the climatic factors (Kumar & Parikh, 2001).

What makes the problem so worrying in India is that it has inadequate arable land but greater dependence of the population on agriculture sector (Mendelsohn et al., 2006) and also have the less adaptive capacity and technological progress to cope with climate change (Birthal et al., 2014). Similarly, effects of climatic variables are anticipated to be severe for Indian agriculture sector because it has large rain-fed area with only about 48 per cent of the cultivated area under assured water supply (Government of India, 2013). Uncertain rainfall and the shortage of irrigation facilities in India are believed to be contributing predominantly to lower crop yields (Pal & Mitra, 2018).

Therefore, agriculture as well as food production systems in India are highly vulnerable to climate change owing to their higher sensitivity to climate change (Dubey & Sharma, 2018). In India, temperature has been elevated by 0.3 to 0.8 ^{o} C per decade during the last few decades (Goswami et al., 2006). The projected climate changes up to 2100 for India show an overall upsurge in air temperature by 2 to 4 ^{o} C combined with a rise in rainfall particularly in the rainy season (Kumar, 2009). Similarly, the rainfall is projected to increase by 6 to 14 per cent at the end of the century (2080) with a high intensity and frequency of precipitation (Chaturvedi et al., 2012). Further, Lobell et al. (2012) revealed that the productivity of wheat crop is extremely sensitive to below 34 °C temperature in northern India.

The IPCC (2007) report found that 0.5 °C elevation in winter-time, minimum temperature is expected to decrease the productivity of wheat in India by 0.45 tons/ha (Easterling et al., 2007). Future climate change could reduce yields in the short-run (2010–2039) by 4–9 per cent and in the long-run (2070–2099) due to lack of adaptation by about 25 per cent (Guiteras, 2009). Interestingly, the projected agriculture output loss by 2100 lies between 10–40 per cent in India

(Aggarwal, 2008). However, these studies established the fact that climate-related factors have significant effects on crop productivity. While the effects of climate change at the state or local level shows identical results with respect to all-India patterns, there are considerable regional differences as well (Saseendran et al., 2000). However, despite its great significance, only a small number of studies has been carried out to evaluate the effects of climatic factors on agricultural output and yield, particularly from the perspectives of developing nations and agriculturally backward regions (Barnwal & Kotani, 2013; Guiteras, 2009). Hence, we aim to evaluate the impact of climate change on crop yields with the focus on India's Telangana state, whose agricultural system is largely dependent on seasonal rainfall. According to few studies (like Padakandla, 2016), Telangana has been one of the region's most vulnerable state's to climate change and environmental hazards.

Climate Change and Agriculture in Telangana

Situated on the southern India, Telangana state is located between latitudes 17° 7′ and 23.4624,'' N and longitudes 79° 12′ and 31.7664'' E. It is India's 12th largest state in terms of both populations (3.52 crores) and geographical area (11,208 lakh hectares) (Government of Telangana, 2016). The study area includes nine undivided districts[1] of Telangana and these districts mainly depend on agriculture. The yearly mean maximum temperature of the state is 42 °C, whereas the mean minimum temperature is about 22–23 °C. Nearly two-thirds of the annual precipitation comes from the south-west monsoon. The northern part of state receives 900–1500 mm annual rainfall, while the southern part of state receives 700–900 mm annual rainfall (Government of Telangana, 2016). It is important to note that more than 50 per cent of the overall cultivated area in Telangana is found to be under the rain-fed agricultural system. This makes the agriculture sector of the state as a rainfall dependent.

As stated above, Telangana state faces various challenges of less rainfall, low ground water levels and drought conditions all of which are directly related to the agricultural output, food security and farmer's livelihood. Given the location of Telangana in the arid and semi-arid climate, agricultural productivity is extremely vulnerable to climate change and drought (Bhavani et al., 2017). Moreover, rural livelihood sector in the state is further exposed to the climatic changes and therefore, it is seen to be at a greater risk than few other Indian states (Revadekar et al., 2012). In the year 2008, World Bank (2008) report confirmed that the adverse impact of climatic change can lead to a substantial reduction in the agricultural income in Telangana region of combined Andhra Pradesh. In recent times, Telangana state has also witnessed high intensive and incidence of droughts and austere farmer suicides (Parida et al., 2018; Tada, 2004).

[1] In the year 2016, Telangana government divided the former 10 districts into 31 new districts. However, this study is based on former 10 districts.

In this background, a well-organised and scientific research of climatic effects on crop production in Telangana state is important for various reasons. This study aims to examine the impact of climate change on mean yield as well as yield variances in major crops (rice, cotton, jowar and groundnut) in three agro-climatic zones of Telangana using district-level panel data during the period of 1956–2015. To the best of our information, this study is the first to empirically examine the climatic impacts on crop yields in Telangana by unequivocally considering likely heterogeneity with respect to crop yield variability (risk) and geographical zones.

The rest of the article is organised as follows. The second section presents a review of the literature. Data sources and methodology are illustrated in the third section. The fourth section provides discussion and analysis of empirical results of climatic effects on major crop yields in Telangana. In the last section, we have presented the conclusion and some policy implications of the study.

A Brief Review on Climate Change and Crop Yields

Over the past two decades, various studies have investigated the effects of climatic factors on crop productivity in various countries of the world. Generally, these studies have estimated the impact of climate change using either (a) crop simulations (agronomic models) or (b) the Ricardian methodology or econometric models. Crop simulation models are based on well-organised scientific experiments where plants are grown in a laboratory or are in the field. Then, simulation experiments are done in different environmental and climate conditions and CO_2 impacts (Saseendran et al., 2000). Crop simulations have the ability to project the possible climatic effects on agricultural production and yield (Sarker et al., 2014).

Some studies that have used the agronomic models to evaluate the effects of climate factors on crop yield include Mearns et al. (1997), Rosenzweig et al. (2002), Saseendran et al. (2000) and Mall and Aggarwal (2002). Mall et al. (2006) have presented an extensive review of crop simulation investigations. Many agronomic modeling studies have revealed the potential adverse consequences of climate change. However, some authors (such as Mendelsohn & Dinar, 1999) have pointed out that agronomic models do not take into account the farmer's adaptation measures and these models can underestimate the positive impacts and overestimate the adverse effects due to climate change.

On the other hand, the Ricardian methodology evaluates the impact of climatic factors on land values or net revenues using the cross-sectional statistics (Mendelsohn et al., 1994; Mendelsohn and Dinar, 1999). Several studies by utilizing Ricardian approach demonstrated that the changes in precipitation and temperature have an adverse effect on land revenue or income. Kumar and Parikh (1998) found that the rice yields decreased by 15–25 per cent and wheat yields by 30–35 percent in India. Further, the net farm income in India decreased by 8 per cent (Mendelsohn et al., 1994). Kumar and Parikh (2001) reported that the anticipated 2 °C upsurge in temperature and 7 per cent upturn in rainfall can diminish farm returns by 9 per cent. Similarly, Sanghi and Mendelsohn (2008) reported that

net revenue of agriculture in India could drop by 4–26 per cent. However, the main limitation of this method is the failure to account for time-independent location-specific factors, namely, the intangible abilities of farmers and soil quality (Barnwal & Kotani, 2013). Also this approach may not be useful for developing nations because of the inefficient land markets.

Furthermore, recent investigations tend to employ panel data methods (Deschenes & Greenstone, 2007). The major studies such as Guiteras (2009), Krishnamurthy (2012), Gupta et al. (2014) and Padakandla (2016) have used the panel data methods and found significant association between climate factors and crops yields. However, these studies have been unable to measure the effects of climate change on yield variance or risk, though the yield variance was found to be affected by climatic factors in several studies (Cabas et al., 2010; Chen et al., 2004; Isik & Devadoss, 2006; Kim & Pang, 2009; Sarker et al., 2014; Sarker et al., 2019).

Existing studies have shown mixed results on the relations of climate change and crop output. We have presented the summary of main findings of few recent empirical studies on climate and crop output nexus in Table 1. Almost all studies have reported that climate change to affect or going to affect adversely the crop output and yield.

Table 1. Summary of empirical investigations on climate change and agricultural yield nexus

Author(s)	Country	Time period	Methodology	Major findings
Deschênes and Greenstone (2007)	USA	1970–2000	Panel approach	Climate factors have a neutral or probably positive effect on crop yields due to farmers adaptation.
Schlenker and Roberts (2008)	USA	1950–2005	Panel approach	Crop yields (corn, cotton and soybeans) are increased slightly up to 29–30 °C temperature and then decreased severely due to higher warming.
Lobell et al. (2011)	Major nations	1980–2008	Panel approach	The global maize and wheat production declined by 3.8 and 5.5 percent, respectively, due to changes in weather.

(Table 1 continued)

(Table 1 continued)

Author(s)	Country	Time period	Methodology	Major findings
McCarl et al. (2008)	USA	1960–2007	Fixed Effects Model	Average crop yields are affected by the mean rainfall and temperature. Higher changes in weather conditions tend to reduce the crop yields.
Mendelsohn et al. (1994)	USA	1982	Ricardian approach	The adverse impact of weather is estimated to lie between 4-6 percent of the value of farm output.
Isik and Devadoss (2006)	Idaho, USA	1939–2001	Panel approach	Rainfall and temperature have a significant adverse effect on wheat, barley, potatoes and sugar beets. Yield variability of wheat and barley is reducing due to climatic factors.
Chen et al. (2004)	USA	1970–2000	Panel approach	Rainfall has positive impact on sorghum and corn yields. Higher temperatures decrease the sorghum yields and its yield variability.
Sarker et al. (2012)	Bangladesh	1972–2009	Time series	Temperature has a significant adverse effect on rice crops (Aus Aman and Boro) but rainfall has a positive impact on three rice crops.
Sarker et al. (2014)	Bangladesh	1972–2009	Panel approach	Temperature is risk-increasing for Aman Aus and Boro rice. Rainfall is risk-increasing for Aman whereas risk-decreasing for Boro and Aus rice.

(Table 1 continued)

(Table 1 continued)

Author(s)	Country	Time period	Methodology	Major findings
Poudel and Kotani (2013)	Nepal	1990–2005	Panel approach	The rice and wheat yields are negatively related with the variation of climate and crop yields more sensitive in low altitude regions of Nepal.
Cabas et al. (2010)	Ontario, Canada	1981–2006	Panel approach	An increase in the variance of rainfall and temperature decrease mean yields and increase yield variability. Yield variability is poorly determined by both seasonal and monthly climate variables.
Kim and Pang (2009)	Korea	1977–2008	Panel approach	The average rice yield is positively associated to temperature and adversely related with rainfall. Both rainfall and temperature variables are risk-increasing factors.
Chen and Chang (2005)	Taiwan	1977–1996	Panel approach	The changes in weather conditions contribute substantially to major crop yield uncertainties. Crop insurance stabilizes revenues of farmers.
Guiteras (2009)	India	1960–1999	Panel approach	The projected weather conditions for the period 2010-2039 shrinks crop yields by 4.5-9 in short-run and 25 percent in long-run (2070-2099).

(Table 1 continued)

(Table 1 continued)

Author(s)	Country	Time period	Methodology	Major findings
Birthal et al. (2014)	India	1969–2005	Fixed Effects Model	Rice, wheat, pigeonpea and chickpea are more susceptible to rise in temperature. However, rainfall has positive impact on major crop yields.
Krishnamurthy (2012)	India	1970–2000	Quantile regression	The study found that a significant drop in wheat output (up to 11%) and moderate reductions in rice output (up to 3%) in India.
Gupta et al. (2014)	India	1966–1999	PCSE	Paddy yield is sensitive to the climatic factors such as rainfall and temperature. However, millets are less affected by climate factors.
Pattanayak and Kumar (2014)	India	1969–2007	Fixed Effects Model	The study found that both day time and night time temperatures unfavourably affect rice yield during different growth stages.
Barnwal and Kotani (2013)	Andhra Pradesh, India	1971–2004	Quantile regression	The kharif rice is more sensitive to temperature and rainfall, while the rabi rice stays mostly resilient to changes in the climate factors.
Saravanakumar (2015)	Tamil Nadu, India	1971–2009	PCSE	The study found a quadratic (inverted U shaped) relationship between (climatic factors and crop yield rice and sorghum).

(Table 1 continued)

(Table 1 continued)

Author(s)	Country	Time period	Methodology	Major findings
Padakandla (2016)	Andhra Pradesh, India	1981–2010	PCSE	Rice, tobacco and jowar yields are adversely impacted by temperature. Cotton and tobacco crop yields are more sensitive to rainfall variability.

Source: Authors' own compilations from available secondary data.

Against this background, the present study examines the effect of climate change on major crop yields and variance in Telangana. We have employed and estimated the Just and Pope production function. This function allows capturing climatic effects on the mean yields and variance (Sarker et al., 2019).

Data and Methodology

Data Sources for District Level Panel Data

This study has used district level panel data (also known as cross-sectional time-series data) for four primary crops, namely, rice, jowar (sorghum), cotton and groundnut from nine former (undivided Andhra Pradesh) districts in Telangana covering the period of 1956–2015. Data on the crop yields and crop-wise area under cultivation of the four principal crops in the state were collected from various issues of Statistical Abstracts of Andhra Pradesh and Statistical Year Books of Telangana, available at Directorate of Economics and Statistics, Government of Telangana. Crop yield is measured in kilograms per hectare {kg/ha} and the crop-wise area is measured in hectare.

We have also collected data of climatic variables from various sources. Statistics of annual and seasonal rainfall for the state and districts were collected from Meteorological Centre Hyderabad and Department of Economics and Statistics, Hyderabad. The mean annual, maximum and minimum temperature data were collected from Indiawaterportal.org but the data were available up to the year of 2002 only at the time of collection. Temperature data since 2002 were collected from various issues of 'Statistical abstracts of Andhra Pradesh'. The panel data on rainfall were for maximum and minimum temperatures on a monthly basis. All the study variables are converted into natural logarithms (Guntukula, 2018). The summary statistics of the study variables are reported in Table 2.

Table 2. Summary Statistics of Study Variables in Telangana

Variables	Observations	Unit	Mean	Std. Dev.	Minimum	Maximum
Rice	540	(kg/ha)	1948.56	772.14	308	4004
MaxT	540	(°C)	32.68	0.68	30.98	35.96
MinT	540	(°C)	21.18	0.84	18.55	25.08
Rainfall	540	(mm)	815.22	232.42	221.82	1773.88
Area	540	(hectare)	128410.4	68320.5	24283	417912
Jowar	540	(kg/ha)	654.1481	271.7066	187	1885
MaxT	540	(°C)	31.59216	.7670315	29.676	35.811
MinT	540	(°C)	20.67293	.9283984	18.167	25.389
Rainfall	540	(mm)	790.6719	228.2241	220.376	1771.704
Area	540	(hectare)	111741.3	88149.83	8	401385
Cotton	540	(kg/ha)	674.13	105.798	40	1380
MaxT	540	(°C)	32.06	0.7189	30.31	35.79
MinT	540	(°C)	20.80	0.8859	18.07	25.61
Rainfall	540	(mm)	800.76	231.0588	220.40	1771.70
Area	540	(hectare)	50351.80	74089.78	12	370363
Groundnut	540	(kg/ha)	987.6037	454.2299	175	2838
MaxT	540	(°C)	31.59216	.7670315	29.676	35.811
MinT	540	(°C)	20.67293	.9283984	18.167	25.389
Rainfall	540	(mm)	790.6719	228.2241	220.376	1771.704
Area	540	(hectare)	32398.74	40202.96	554	208802

Source: Authors' own calculations.

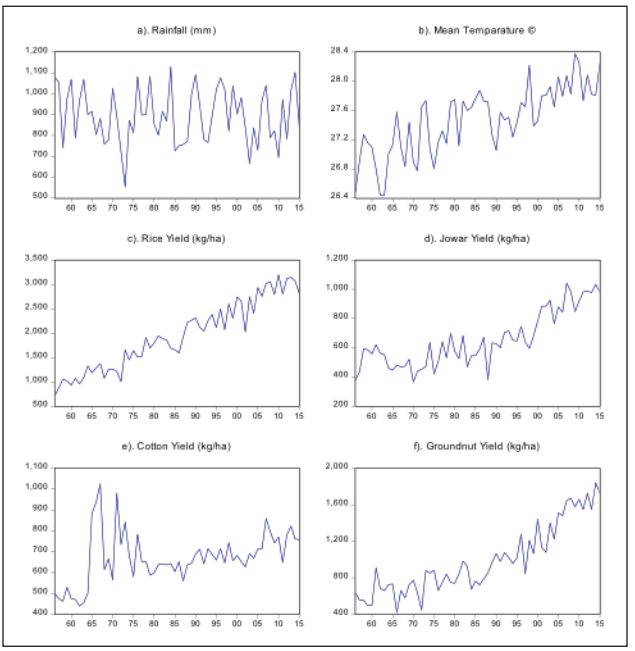
Table 3. Agro-climatic (A-C) Zones of Telangana and Their Characteristics

A-C zones	Districts	Total Area	Annual Rainfall	Soil type	Major crops
Northern Zone	Karimnagar, Adilabad, Nizamabad	35.5 (lakh ha)	Normal rainfall 900-1150 (mm)	Red earths with loamy soils (Chalkas) and Black Cotton soils	Rice, Maize, Cotton, Jowar, Groundnut, Redgram, Soybean
Central Zone	Warangal, Medak, Khammam	30.6 (lakh ha)	Normal rainfall 800-1150 (mm)	Red earths with loamy soils (Chalkas), Black Cotton soils and Red sandy soils	Rice, Cotton, Maize, Groundnut, Greengram, Jowar, Chilli
Southern Zone	Nalgonda, Mahabubnagar, Ranga Reddy	39.3 (lakh ha)	Normal rainfall 500-670 (mm)	Red soils, Chalkas	Cotton, Rice, Redgram, Maize, Groundnut, Jowar, Greengram

Source: Compiled from the Drought Manual, Telangana, 2016.

The study has incorporated undivided 9 districts of Telangana in which agricultural sector is dominant. These districts are Adilabad, Karimnagar, Nizamabad, Warangal, Khammam, Medak, Nalgonda, Rangareddy and Mahabubnagar. Hyderabad district area has been excluded from this investigation because it has insignificant crop cultivation area. According to the directorate of economics and statistics (DES) of Telangana, the state has been categorised into three agro-climatic (AC) zones. Table 3 shows information about three agro-climatic zones along with the districts in respective zone, climatic conditions, soil types, major crops and percentage share of the agro-climate zones in overall geographical area of state. To examine the impact of climate change on yields and yield variance, we have selected four major food and non-food crops, namely, rice, jowar, cotton and groundnut. The main reason behind selecting these crops is that they are principal crops together account to about 60 per cent of the overall gross cropped area in Telangana. So, the impact of climatic factors on these crop yields can have significant implications on food security, income generation and livelihood of the farm households in the state. Secondly, consistent and comparable time-series crop-wise data are not available for other crops at the district-level. Trends of the climatic factors such as rainfall and mean temperature, along with trends of the yields of study crops during the study period are presented in Figure 1 (a to f). It can be seen that the rainfall has more fluctuations among the study variables. The mean temperature has shown an increasing trend over the period. The crop yields also demonstrated an increasing trend in productivity over the period. These trends are more prominent for rice, jowar and groundnut compared to cotton yield.

Figure 1. Climate Variables and Crop Yield trends in Telangana 1956–2015



Source: Authors' own compilations from available secondary data.

Methodology

Model Specification

As already stated, we have applied and estimated the production function developed by Just and Pope (1978). Literature review shows that significant portion of empirical studies has utilised either a crop simulation or a cross-sectional model. The crop simulation model using 'general circulation models' is found to perform inadequately (Schlenker & Roberts, 2008). Moreover, the extensive use of 'Ricardian approach' is not capable to capture the impact of omitted variables on crop output which produces biased results (Deschenes & Greenstone, 2007). In order to overcome this omitted variable issue, the production function developed by Just and Pope (1978) has been successfully utilised by many studies (e.g., Chen et al., 2004; Isik & Devadoss, 2006, Kim & Pang 2009; Poudel et al., 2014: Sarker et al., 2019).

In the present study, in order to evaluate the impacts of climatic factors on the mean level of the yield and variability of the crop yields, the stochastic production function approach introduced and elaborated by Just and Pope (1978; 1979) is employed. The basic concept underpinning this method is that a stochastic production function can be decomposed into two parts: the first section is related to the mean level of productivity whereas the second section is linked with the vari-

ability of that productivity (Cabas et al., 2010; Kim & Pang, 2009; Sarker et al., 2014). The general formula of the Just and Pope production function (Just & Pope, 1978) is as follows:

$$y = f(X) + h(X)\epsilon \tag{1}$$

where is the crop yield and is the vector of explanatory variables. ε is the random error with mean zero and variance (σ^2). The parameter assessment of $f(X)$ produces the mean impact of the independent variables on the crop yield whereas $h(X)$ provides their influence on the variability of the crop yield (Chen & Chang, 2005). Based on Chen et al. (2004) and Sarker et al. (2014) the production function of following form is assessed as follows:

$$y = f(x) + u = f(X,\beta) + h(X,\alpha)\epsilon \tag{2}$$

where y is crop yield (rice, jowar, cotton and groundnut), X is the vector of explanatory variables (e.g., maximum and minimum temperatures, rainfall, area, time period and location) and ε is the exogenous output shock with $E(\epsilon) = 0$ and $Var(\epsilon) = \delta_{\epsilon}^2$. Using this function, independent variables affect both the average and variability of crop yield because $E(y) = f(x)$ and $Var(y) = Var(u) = h(.)$. The factor assessment of $f(.)$ provides the average impacts of the independent variables on yield, whereas $h(.)$ gives the effects of the covariates on the variability of crop yield (Isik & Devadoss, 2006). It is very important to note that the positive sign on any factor of $h(.)$ infers an increase in that variable, that is, an upsurge of the variability of the crop yield. A negative sign on the variability function implies a reduction of the yield variability demonstrating that climatic variables are risk decreasing inputs.

Generally, three functional forms of production functions are used to estimate Just and Pope production function—Cobb–Douglas, quadratic and translog production functions (Chen et al., 2004; Kim & Pang 2009; Sarker et al., 2014). Due to the multiplicative interaction between the variance and mean, the translog functional form can violate the Just and Pope assumptions (Tveteras & Wan, 2000). Therefore, purposively, the linear Cobb–Douglas form has been used for the mean and variability of the crop yield function. This functional form is consistent with the Just and Pope (1979) assumptions as argued by Kim and Pang (2009).

Estimation Method

All the parameters in the Equation (2) can be estimated by employing the maximum likelihood estimation (MLE) method suggested by Saha et al. (1997) or three-step feasible generalised least squares (FGLS) method proposed by Just and Pope (1979). Most of the empirical studies have employed the FGLS estimation procedure but MLE is more consistent and efficient estimator than FLGS particularly under a small sample situation (Saha et al., 1997). Since we have a large sample size, this study has used the three-step FGLS estimation procedure as explained in Judge et al. (1985). Moreover, panel data model estimation is generally comprised of both cross-section and time series data and can solve the problems of auto-correlation and heteroscedasticity (Gujrati 2004; Hill et al., 2008). However, these two problems are addressed in a better way using the FGLS procedure as it assumes that panels are not auto-correlated and homoscedastic

(Wooldridge, 2002). Generally, the fixed effects (FE) and random effects (RE) models are designed to estimate the panel data model (Baltagi, 2005). This study selected fixed effects model based on Hausman test results and this preference of fixed effects model is consistent with few existing studies (Barnwal & Kotani, 2010; Cabas et al., 2010; Guntukula & Goyari, 2020; Kim & Pang, 2009; Sarker et al., 2014).

Formulation of Econometric Model

As mentioned earlier, the major disadvantage of the 'Ricardian approach' lies in its lack of ability to include omitted variables such as soil quality and unobservable skills of the crop growers which are also known as location specific and time-independent factors (Barnwal & Kotani, 2010). In order to overcome this omitted variable issue, we have incorporated agro-climatic zone dummies in the model. However, regional agro-climatic zone dummies are included only in the average yield function but not to the yield variance function. This is based on the assumption that different agro-climate zones have different average yields with almost similar yield variabilities across zones (Sarker et al., 2014). Linear trend was incorporated to include the impact of technological change over time. We have estimated the impact of climate change on crop yields and variability by specifying three models (Model I, Model II and Model III. In model I, we have included all the explanatory variables such as rainfall, area, maximum and minimum temperatures. However, maximum and minimum temperatures are not mutually exclusive in nature. Hence, instead of taking both maximum and minimum temperatures in one model, we have framed the model II and model III independently using minimum and maximum temperatures separately.

Model I: The econometric specification of mean yield function $f(X,\beta)$ is expressed as follows:

$$Y_{it} = \beta_0 + \beta_1 MxT_{it} + \beta_2 MnT_{it} + \beta_3 Rf_{it} + \beta_4 A_{it} + \beta_5 T_{it} + \sum_{l=1}^{L-1} \alpha_l D_l + e_{it} \quad (3)$$

where Y_{it} is the crop yields (kg/ha) for production district i and time t; MxT_{it}, MnT_{it}, Rf_{it} and A_{it} are the maximum temperature, minimum temperature, rainfall and area under crop, respectively; $\alpha_l D_l$ (i = 1, 2…) are agro-climatic zonal dummies taking values 1 and 0. T (year) is time trend variable to capture impacts of developments in technology. Further, the yield variability function $h(X,\alpha)$ is given as follows:

$$\ln e_{it}^2 = \alpha_0 + \alpha_1 MxT_{it} + \alpha_2 MnT_{it} + \alpha_3 Rf_{it} + \alpha_4 A_{it} + \alpha_5 T_{it} + \mu_{it} \quad (4)$$

where $\ln e_{it}^2$ log of squared residuals from first step ordinary least squares (OLS) is a dependent variable and explanatory variables are maximum and minimum temperatures, rainfall, area and trend. Further, α's are the parameter estimates and other independent variables are as defined in Equation (3). The estimation of the model II incorporates all the study variables excluding minimum temperature. On the contrary, Model III incorporates all the study variables excluding maximum temperature.

Panel Unit Root Test

To avoid possible spurious regressions, it is essential to perform panel unit root test for stationarity for each study variable before estimating the model (Chen & Chang, 2005; Poudel & Kotani, 2013). Moreover, the pre-condition of Just and Pope production function is that the study variables in the model must be stationary (Chen et al., 2004). There are several types of tests for this purpose. Some of these include tests developed by Levin, Lin and Chu (LLC) (2002), Im et al. (2003), Breitung (2001) and Fisher-type test of Maddala and Wu (1999). In this study, we have employed Fisher-type test using Augmented Dicky-Fuller (ADF) test because this test gives more accurate results (Baltagi, 2005).

Fisher-type panel unit root test: Maddala and Wu (1999) and then Choi (2001) offered the Fisher-type unit root test, which pools the p-values from individual unit root tests. If is well-defined as the p-value from any individual test for cross-section section *i*, under the null of unit root test, then, for all N cross-sections, the asymptotic results are written as follows:

$$-2\sum_{i=1}^{N} ln(\pi_i) \rightarrow \chi^2 2N \tag{5}$$

The main advantage of the Fisher-type test is that it can be employed for various lag lengths in the individual ADF-regressions (Baltagi, 2005). Moreover, this test, using ADF-regressions with bootstrap-based values, makes the best and as a result is the most preferred method for testing the existence of non-stationary in the panel data (Maddala & Wu, 1999). The null and alternative hypotheses for the ADF test are: (a) Null hypothesis (Ho): Entire panels contain unit roots and (b) Alternative hypothesis (Ha): At least one panel is stationary.

Results and Discussions

Pair-wise Correlation and Panel Unit Root Test Results

The pair-wise correlations among the study variables are reported in Table 4. Results are mixed. The rice yield is positively and significantly correlated with maximum temperature, minimum temperature and rainfall. Moreover, jowar yield is positively correlated with rainfall, minimum and maximum temperatures but minimum temperature and rainfall variables are not statistically significant at 5 per cent level. Cotton yield is positively and significantly correlated with minimum temperature only. Cotton yield is also positively correlated with maximum temperature but statistically insignificant at 5 per cent level. The cotton yield is also negatively correlated with rainfall. Lastly, groundnut yield is positively correlated with maximum temperature, minimum temperature and rainfall and correlation coefficients are statistically significant at 5 per cent level.

Table 4. Pair-wise Correlation Matrix of Study Variables in Telangana

	Rice yield	**Max. Temp.**	**Min. Temp.**	**Rainfall**	**Area**
Rice yield	1.0000				
Max. Temp.	0.4040*	1.0000			
Min. Temp.	0.1967*	0.6293*	1.0000		
Rainfall	0.1563*	–0.1454*	0.0146	1.0000	
Area	–0.2110*	0.1321*	0.1981*	–0.3245*	1.0000
	Jowar yield	Max. Temp.	Min. Temp.	Rainfall	Area
Jowar yield	1.0000				
Max. Temp.	0.2528*	1.0000			
Min. Temp.	0.0239	0.6293*	1.0000		
Rainfall	0.0644	–0.1454*	0.0146	1.0000	
Area	–0.4530*	–0.2133*	–0.1327*	–0.1583*	1.0000
	Cotton yield	Max. Temp.	Min. Temp.	Rainfall	Area
Cotton yield	1.0000				
Max. Temp.	0.0446	1.0000			
Min. Temp.	0.1685*	0.6001*	1.0000		
Rainfall	–0.0625	–0.1281*	0.0264	1.0000	
Area	–0.0682	0.5118*	0.2358*	0.1460*	1.0000
	Groundnut yield	Max. Temp.	Min. Temp.	Rainfall	Area
Groundnut yield	1.0000				
Max. Temp.	0.4040*	1.0000			
Min. Temp.	0.1967*	0.6293*	1.0000		
Rainfall	0.1563*	–0.1454*	0.0146	1.0000	
Area	–0.2110*	0.1321*	0.1981*	–0.3245*	1.0000

Source: Authors' own calculations.

Note: * indicates the statistical significance at 5% level.

Table 5. Fisher Type Panel Unit Root Test Results of Climate Variables in Telangana

Crops	Variables	Fisher test (ADF) Test statistics	
		Without Trend	With Trend
Rice	Yield	53.49 ***	272.60 ***
	Maximum Temperature	133.97 ***	239.45 ***
	Minimum Temperature	144.88 ***	206.93 ***
	Rainfall	503.53 ***	466.61 ***
	Area	161.03 ***	217.33 ***
Jowar	Yield	129.58 ***	231.45 ***
	Maximum Temperature	142.33 ***	236.99 ***
	Minimum Temperature	157.22 ***	206.28 ***
	Rainfall	512.48 ***	477.05 ***
	Area	134.23 ***	184.94 ***
Cotton	Yield	52.66 ***	38.15 ***
	Maximum Temperature	139.66 ***	245.90 ***
	Minimum Temperature	151.62 ***	212.89 ***
	Rainfall	510.14 ***	473.10 ***
	Area	35.99 ***	35. 91 ***
Groundnut	Yield	65.04 ***	253.61 ***
	Maximum Temperature	142.33 ***	236.99 ***
	Minimum Temperature	157.22 ***	206.28 ***
	Rainfall	512.48 ***	477.05 ***
	Area	161.81 ***	174.59 ***

Source: Authors' estimations.

Note: *** $p < 0.01$.

Prior to estimating the Just and Pope production function, Fisher-type (ADF) panel unit root test is applied to check the stationarity properties of study variables. The panel unit root test results are presented in Table 5. The test statistics of the ADF regression for the crop yields (rice, cotton, jowar and groundnut) and climatic variables (temperature and rainfall) are found to identical in both models with trend and without trend. The panel unit root test results indicate that all study variables are statically significant and the null hypothesis of unit roots is rejected at the 1 per cent level of significance. This infers that the study variables are stationary at the level or I (0).

The panel unit root test results are consistent with other studies such as Kim and Pang (2009), Sarker et al. (2019) and Sarker et al. (2014). Therefore, Just and Pope production function using the three-step FGLS estimation procedure can be applied to district-level panel data without any differencing.

Table 6. Estimated Results for Rice in Telangana

Rice	Model I		Model II		Model III	
Mean yield function *f(x)*	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value
Trend	0.016***	0.000	0.093***	0.000	0.097***	0.000
Maximum Temperature	–1.907***	0.003	–1.502***	0.003		
Minimum Temperature	0.589*	0.072			0.039	0.881
Rainfall	0.540***	0.000	0.552***	0.000	0.595***	0.000
Area	1.008***	0.000	1.005***	0.000	1.076***	0.000
Northern AC Zone	20.742***	0.000	21.210***	0.000	16.561***	0.000
Central AC Zone	20.695***	0.000	21.166***	0.000	16.527***	0.000
Southern AC Zone	20.804***	0.000	21.274***	0.000	16.634***	0.000
Constant	3.548***	0.000	0.811***	0.000	–3.827***	0.000
Yield variability function *h(x)*						
Trend	0.005***	0.000	0.005***	0.000	0.005***	0.000
Maximum Temperature	–0.050***	0.000	–0.027***	0.001		
Minimum Temperature	0.038***	0.000			0.023***	0.000
Rainfall	0.028***	0.000	0.029***	0.000	0.029***	0.000
Area	0.061***	0.000	0.061***	0.000	0.061***	0.000
Constant	3.010***	0.000	3.049***	0.000	2.878***	0.000
Model summary						
Log likelihood	1212.641		1212.557		1210.81	
Wald Chi-square	749489.67		755748.94		729741.91	
Prob > Chi-square	0.0000		0.0000		0.0000	

Source: Authors' own estimations.

Note: *** $p < 0.01$ and * $p < 0.10$.

Crop-wise Estimation Results

The present section deals with crop-wise empirical results and discussions. The effect of climate related variables on crop yields in Telangana has been examined under the panel data approach with 540 observations and 9 panel groups (i.e., annual data for 60 years during 1956–2015 for 9 districts). The estimations of the mean yield and variability functions have been done using the three-step feasible generalised least squares (FGLS) estimation procedure.

1. **Estimated Results for Rice:** The estimated results for rice are shown in Table 6. The estimated coefficients of explanatory variables for three rice models are collectively statistically significant since the aggregate Wald Chi-square statistic has a p-value of 0.000. In the mean yield function of rice, all the study variables, including agro-climatic zone dummies, are significantly influencing rice yield over the study period. Rainfall has a significant influence on average rice yield in Telangana as per results in three models. This infers that the rice cultivation in Telangana largely depends on rainfall. However, the maximum temperature is adversely associated with the rice yield in model I and model II and it is statistically significant at 1 per cent level of significance. This indicates that the growing maximum temperature may reduce the rice yield in Telangana. Moreover, minimum temperature has a significant positive impact on rice yield in model I and model III. The positive effect of minimum temperature may not be able to counterbalance the adverse effect of maximum temperature as observed in coefficient values. Our results are almost similar to other studies such as Padakandla (2016) and Pattanayak and Kumar (2014). An increase in area under rice has a positive and significant effect on yield in three rice models. The time-trend variable is positively associated with the mean yield in three models. This infers that technological progress has a positive effect on rice yield over time. Lastly, three agro-climatic zone variables have significant and positive influences on rice yield in three models. However, the magnitude of influence varies in three rice models. Similar empirical results were obtained in studies of other regions such as Poudel and Kotani (2013) and Sarker et al. (2014).

The results of yield variability function for rice crop are also reported in Table 6. It can be observed that climatic factors, including area under rice and trend variable, are statistically significant at 1 per cent level in the rice variability function. The maximum temperature has an adverse impact on the yield variance of rice. This indicates that maximum temperature is a risk-reducing factor for rice yield variability in two models. The other climatic factors such as rainfall and minimum temperature are positively influencing the yield variability of rice. This implies that the rainfall and minimum temperature are risk-enhancing factors in rice yield variance function. Similarly, an increase in cultivation area under rice crop has a positive effect on yield variability of rice in three models. This suggests that increased area under rice is a risk-enhancing factor. Interestingly, the time trend has a significant favourable effect on rice yield variance. This can be interpreted in such a way that rice yield variability possibly may be increasing over time due to technological advancements such as new HYVs seeds, irrigation and better use of fertilizers.

Table 7. Estimated Results for Jowar in Telangana

Jowar	**Model I**		**Model II**		**Model III**	
Mean yield function *f(x)*	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value
Trend	–0.302***	0.000	–0.335***	0.000	–0.289***	0.000
Maximum Temperature	5.939***	0.000	0.336	0.645		
Minimum Temperature	–7.203***	0.000			–4.693***	0.000
Rainfall	0.241***	0.000	0.189***	0.001	0.122**	0.024
Area	0.304***	0.000	0.328***	0.000	0.267***	0.000
Northern AC Zone	–153.344***	0.000	–171.125***	0.000	–133.457***	0.000
Central AC Zone	–153.399***	0.000	–171.208***	0.000	–133.509***	0.000
Southern AC Zone	–153.583***	0.000	–171.386***	0.000	–133.690***	0.000
Constant	25.840***	0.000	28.408***	0.000	24.874***	0.000
Yield variability function *h(x)*						
Trend	0.003***	0.000	0.003***	0.000	0.003***	0.000
Maximum Temperature	–0.076***	0.000	–0.019***	0.000		
Minimum Temperature	0.078***	0.000			0.048***	0.000
Rainfall	–0.002***	0.000	–0.002***	0.000	–0.001***	0.000
Area	–0.003***	0.000	–0.003***	0.000	–0.002***	0.000
Constant	3.671***	0.000	3.704***	0.000	3.484***	0.000
Model summary						
Log likelihood	868.6839		869.2676		867.954	
Wald Chi-square	1.76e+06		1.72e+06		1.70e+06	
Prob > Chi-square	0.0000		0.0000		0.0000	

Source: Authors' own estimations.

Note: *** $p < 0.01$ and ** $p < 0.05$.

2. **Estimated Results for Jowar (Sorghum):** The estimated results for jowar are provided in Table 7. Estimated coefficients of all explanatory variables for three jowar models are collectively statistically significant since the aggregate Wald Chi-square statistic has a p-value of 0.000. The explanatory variables such as rainfall, maximum and minimum temperatures, area and agro-climatic zone dummies are statistically significant in three jowar models. However, maximum temperature in model II is not statistically significant in jowar mean yield function. Rainfall has a significant positive effect on mean yields in three jowar models. Jowar is a rain-fed crop in Telangana and due to this reason, possibly the effect of rainfall is favourable. Likewise, maximum temperature has a significant positive effect on mean yields in model I. Jowar is considered to be mainly a heat resistant crop and hence we can say that the upper temperature may not affect the crop yield to a great extent. However, minimum temperature has an adverse effect on mean yield of jowar in two models. The positive impact of maximum temperature may be unable to offset the adverse effect of minimum temperature as observed from estimated coefficient values. Our findings of minimum and maximum temperature are contrary to findings by Padakandla (2016). An increase in area under jowar crop has a positive impact on jowar yield. Further, time-trend is adversely associated with the mean yield of jowar. All regional agro-climatic zone variables in three models are found to be statistically significant and have adverse effects on jowar mean yields in Telangana.

The yield variability function results of the jowar crop are also shown in Table 7. Rainfall, area, minimum and maximum temperatures are statistically significant in yield variability function. The minimum temperature has a significant and positive effect on jowar yield variability. This implies that the minimum temperature in Telangana is a risk-enhancing factor for yield variance of jowar. However, maximum temperature and rainfall have adverse impact on jowar yield variance. This indicates that the rainfall and maximum temperature are risk-reducing factors for jowar yield. Similarly, an increase in area under jowar crop has a significant adverse effect on yield variance of jowar. The increased cropped area is also a risk-reducing factor in Telangana. Lastly, the time-trend factor has a significant favourable effect on jowar yield variability. Therefore, we can conclude that the trend variable is a risk-enhancing factor for jowar crop. These results of mean yield and variability function results for jowar are almost similar to few studies like Boubacar (2010).

3. **Estimated Results for Cotton:** Cotton is one of the important commercial crops in Telangana and is mostly cultivated under rain-fed conditions. The empirical results of cotton crop are shown in Table 8. Estimated results of explanatory variables for three cotton models are collectively statistically significant since the aggregate Wald Chi-square statistic has a p-value of 0.000. All the climatic variables, cotton area and agro-climatic zone variables are statistically significant in three cotton models. Rainfall has a significant adverse effect on mean cotton yield in Telangana. As mentioned

earlier, cotton is a rain-fed crop, an increase in rainfall may tend to decrease the cotton yield. Likewise, maximum temperature is adversely related to mean yield of cotton. The positive impact of minimum temperature on cotton yield may be unable offset the adverse effect of maximum temperature as reflected in estimated results. Present results of minimum and maximum temperature in mean yield function of cotton are contrary to some studies such as Padakandla (2016) and Guntukula (2020). The empirical outcomes of climatic factors in mean yield function of cotton are similar with Kumar et al. (2015). The trend factor is positively related to cotton yield. This implies that the technological progress in Telangana has a positive effect on cotton yield. All the regional agro-climatic zone variables such as southern, central and northern zones are significant and these agro-climatic zones have the favourable effect on mean cotton yield.

The estimated coefficients of yield variability function for the cotton crop are given in Table 8. The mean minimum temperature has a substantial favourable effect on the yield variance of cotton. In other words, the minimum temperature is a risk-enhancing factor for cotton in Telangana. However, maximum temperature has a significant negative effect on the yield variance of cotton. This implies that the maximum temperature is a risk-reducing factor for cotton. Similarly, rainfall has a substantial adverse effect on the yield variance of cotton. We can say that the rainfall is a risk-reducing factor for cotton yield variance. Further, area under cotton is positively related to yield variability of cotton in three models. This indicates that the area is a risk-enhancing factor for cotton. Furthermore, trend is positively related to yield variability of cotton in the model I and model II, but it is negatively related to the yield variance of cotton. This implies that the technological progress in Telangana is a risk-enhancing factor for cotton in Telangana. The yield variability findings of our study are, to some extent, similar to few studies like Kumar et al. (2015). We find some interesting results for cotton crop. Variables such as rainfall and maximum temperature, which have an adverse impact on mean cotton yield, are the risk-reducing factors for cotton yield variance. Variables like minimum temperature, which have a positive effect on mean cotton yield, are risk-enhancing factors for cotton yield variance.

4. **Estimated Results for Groundnut:** The estimated results of groundnut crop are reported in Table 9. The estimated coefficients of explanatory variables for three groundnut crop models are collectively statistically significant since the aggregate Wald Chi-square statistic has a p-value of 0.000. All the study variables, including agro-climatic zone dummy variables are significantly influencing groundnut yields over the study period in the mean yield function. However, the mean maximum temperature is not statistically significant in model I.

Table 8. Estimated Results for Cotton in Telangana

Cotton	Model I		Model II		Model III	
Mean yield function *f(x)*	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value
Trend	0.083***	0.000	0.052*	0.084	–0.068***	0.000
Maximum Temperature	–283.787***	0.000	–74.314*	0.086		
Minimum Temperature	128.893***	0.000			113.266***	0.000
Rainfall	–12.362***	0.000	–5.262*	0.084	–19.025***	0.000
Area	1.200***	0.000	0.550*	0.086	1.985***	0.000
Northern AC Zone	780.082***	0.000	352.967*	0.085	49.955***	0.000
Central AC Zone	780.169***	0.000	353.310*	0.085	50.123***	0.000
Southern AC Zone	779.839***	0.000	352.883*	0.086	49.808***	0.000
Constant	16.580***	0.000	–8.955	0.120	8.685***	0.000
Yield variability function *h(x)*						
Trend	0.002***	0.000	0.001***	0.000	–0.001***	0.000
Maximum Temperature	–4.827***	0.000	–2.263***	0.000		
Minimum Temperature	2.165***	0.000			0.796***	0.000
Rainfall	–0.211***	0.000	–0.158***	0.000	–0.134***	0.000
Area	0.023***	0.000	0.016***	0.000	0.141***	0.000
Constant	15.159***	0.000	12.510***	0.000	2.174***	0.000
Model summary						
Log likelihood	425.3464		387.0905		394.1517	
Wald Chi-square	362494.10		701337.54		729739.81	
Prob > Chi-square	0.0000		0.0000		0.0000	

Source: Authors' own estimations.

Note: *** $p < 0.01$ and ** $p < 0.10$.

Table 9. Estimated Results for Groundnut in Telangana

Groundnut	**Model I**		**Model II**		**Model III**	
Mean yield function *f(x)*	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value
Trend	–0.084***	0.003	–0.069**	0.022	0.027***	0.015
Maximum Temperature	–0.913	0.277	5.392**	0.026		
Minimum Temperature	8.975***	0.000			7.627***	0.004
Rainfall	–1.054***	0.005	–0.819**	0.032	–0.781**	0.027
Area	0.092***	0.001	0.111***	0.004	0.098***	0.002
Northern AC Zone	–42.956***	0.003	–35.564**	0.021	–38.326***	0.006
Central AC Zone	–42.956***	0.003	–35.527**	0.021	–38.274***	0.006
Southern AC Zone	–43.007***	0.003	–35.616**	0.021	–38.358***	0.006
Constant	1.395***	0.000	4.450***	0.003	4.160***	0.002
Yield variability function *h(x)*						
Trend	0.006***	0.000	0.005***	0.000	0.006***	0.000
Maximum Temperature	–0.062***	0.000	–0.472***	0.000		
Minimum Temperature	–0.519***	0.000			–0.583***	0.000
Rainfall	0.078***	0.000	0.073***	0.000	0.078***	0.000
Area	–0.004***	0.000	–0.006***	0.000	–0.006***	0.000
Constant	4.956***	0.000	4.860***	0.000	4.947***	0.000
Model summary						
Log likelihood	957.7352		954.6335		955.2668	
Wald Chi-square	881239.67		937163.39		793470.73	
Prob > Chi-square	0.0000		0.0000		0.0000	

Source: Authors' own estimations.

Note: *** $p < 0.01$ and ** $p < 0.05$.

Rainfall has been found to have negative effect on groundnut yield in Telangana. However, the minimum temperature is positively related to groundnut yield. Maximum temperature has an adverse effect on groundnut yield in model I, but it is not statistically significant. In model II, maximum temperature has a significant adverse effect on mean yield. In model I, the adverse effect of rainfall and maximum temperature on groundnut yield is offset by the positive effect of minimum temperature. The cropped area of groundnut has a substantial favourable effect on mean yield in three groundnut models. The time trend variable has a significant adverse effect on groundnut yield in the model I and model II. However, we found that the trend is positively related to mean yield in model III. Further, all the three agro-climatic zone variables in the mean yield function are statistically significant and related negatively to groundnut yield. The empirical findings of groundnut are almost similar to few studies like Kumar et al. (2015) and are contrary to studies like Padakandla (2016).

The yield variability function results of the groundnut crop are also reported in Table 9. Rainfall, area, minimum and maximum temperatures are significantly influencing the variability of groundnut yield in Telangana. Rainfall has a positive effect on yield variance. In other words, rainfall is a risk-enhancing factor for groundnut. However, maximum and minimum temperatures have significantly adverse effect on the yield variance of groundnut. This indicates that the maximum and minimum temperatures are risk-reducing factors for groundnut. Similarly, cropped area is negatively related to the yield variance of groundnut. This implies that the area is the risk-reducing factor for groundnut. Furthermore, trend has a significant positive effect on yield variance. This implies that technological advancement is increasing the yield variability of the crop in the state during the study period. Our results on climatic factors in yield variability function of groundnut are almost contrary to studies like Kumar et al. (2015).

Conclusions and Policy Implications

In this article, we have empirically examined the effects of climatic factors on major crop yield and its variability by applying the Just and Pope production function in Telangana over the period of 1956–2015. This study has used district level panel data on four crop yields, namely rice, jowar, cotton and groundnut and climatic variables such as rainfall, maximum temperature and minimum temperature. For empirical analysis, we have employed the three-stage feasible generalised least squares estimation technique.

The empirical results from the estimated econometric models have shown that the effects of the rainfall, minimum temperature and maximum temperature on major crop yields vary across different crops. However, the empirical findings have revealed that climatic factors such as rainfall and temperature have adverse effects on crop productivity in general. Maximum temperature has a significantly adverse effect on rice, cotton and groundnut yields whereas favourable effect on jowar yields. Given that jowar (Sorghum) is the stress tolerant crop, it is possible

that an increase in maximum temperature may not affect the jowar yield to a great extent. Some studies have shown an adverse influence of the upsurge in maximum temperature on rice productivity in India (Mall et al., 2006; Pattanayak & Kumar, 2014). However, the minimum temperature has a significantly adverse effect on jowar yield while advantageous effect on rice, cotton and groundnut. Rainfall has a significantly adverse effect on cotton and groundnut yield whereas positive impact on rice and jowar yields. We have found that the crop area has a significantly positive effect on all crop yields under study. In other words, an increase in area under the study crops is positively influencing crop yield. Agro-climatic zone variables have mixed results, positive on the rice and cotton but adverse effect on jowar and groundnut yields.

We have also estimated the impact of climatic factors on yield variability of four primary crops. The empirical findings have revealed that the climatic factors have significant influences on the variability of major crop yields in Telangana. Maximum temperature is a risk-reducing factor for all the study crops. However, minimum temperature is a risk-enhancing factor for rice, jowar and cotton. The impact of rainfall on yield variance is negative for jowar and cotton and positive for rice and groundnut. This implies that rainfall is risk-reducing factor for jowar and cotton while risk-enhancing factor for rice and groundnut. The results of the study have revealed that different crops are influenced differently by the climatic factors in Telangana. Therefore, results from the present study have important policy implications for developing sub-national level adaptation strategies for farm households in Telangana. Since climate variables are beyond the control of farm households and changes in climate variables affect crop yields, there must be proper adaptation policies for both short run and long run. One measure is to implement crop insurance policy as an adaptation practice to diminish the possible monetary losses to the crop growers. The government also must support the development of new varieties crop seeds which are temperature (heat or cold) tolerant for sustainable agricultural growth.

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The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

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